Class

- 3/17/2026 St Patricks Day
- Office Hours Tomorrow 11am
- Takehome Mid/HW 4 due NEXT THURS
- Reading Start into Ch 8 Griff + ~last Field Theory reading
- TODAY → DIRAC EQ. & QED 1






Midterm Assignment Help

- Problem 2:
  - Really the problem is about proving that the transformation β_{CM} to transform to the CM is essentially calculated as p/E but using the sum of the pa pb 4-v's for the p and E in beta: ie what you would by assuming that the frame reference is a boosted single particle with the 4-mom equal to the sum of the two lab frame p's, pa,pb.

$$S^\mu = p_a^\mu + p_b^\mu$$

prove

$$\beta_{cm} = \frac{|S|}{|E_S|} = \frac{S_z}{E_S}$$

b.c. beam dir p_T's all 0

$|S| = S_z$

  - and you may assume that you could just use that $\beta \equiv \beta_{\|}$ w/ e.g. 1+beta/1-beta [or coshy/sinhy...] vers of formulae to calculate y - in other words, you can use that to show what you are supposed to which is of course:

$$y = \frac{1}{2} \ln\left(\frac{S_0 + S_z}{S_0 - S_z}\right)$$

my point is that there may be at least a tiny bit of extra logic to jump from that form of βcm and assuming that that implies also that y cm would have the corresponding form – can you really use a sum as if it is a single particle p_mu for y ? — but you do not have to include/address this jump (if you don't want to)

You CAN and should try to show from the direct transformation as described in the problem but but I won't take much off if you don't, and further just assume that such a boost beta can define the ycm

QUICK REVIEW IN TOY THEORY

- Indentification of the Mendelstaam variables

Example Toy (ABC) Theory Feynman Rules

- Feunman Rules to Calculate M

Our problem is to find the amplitude $\mathcal{M}$ associated with a given Feynman diagram. The ritual is as follows[5]:

1. *Notation*: Label the incoming and outgoing four-momenta $p_1, p_2, \ldots, p_n$ (Figure 6.6). Label the internal momenta q_1, $q_2, \ldots$. Put an arrow beside each line, to keep track of the 'positive' direction (forward in time for external lines, arbitrary for internal lines).
2. *Vertex factors*: For each vertex, write down a factor

$$-ig$$

g is called the *coupling constant*; it specifies the strength of the interaction between A, B, and C. In this toy theory, g has the dimensions of momentum; in the 'real-world' theories, we shall encounter later on, the coupling constant is always dimensionless.

3. *Propagators*: For each internal line, write a factor

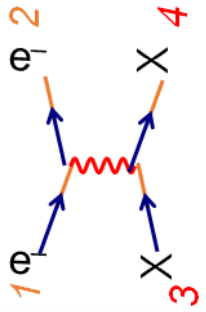
$$\frac{i}{q_j^2 - m_j^2 c^2}$$

where q_j is the four-momentum of the line and m_j is the mass of the particle the line describes. (Note that $q_j^2 \neq m_j^2 c^2$, because a virtual particle does not lie on its mass shell.)

4. *Conservation of energy and momentum*: For each vertex, write a delta function of the form

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

where the k's are the three four-momenta coming *into* the vertex (if the arrow leads outward, then k is *minus* the four-momentum of that line). This factor imposes conservation of energy and momentum at each vertex, since the delta function is zero unless the sum of the incoming momenta equals the sum of the outgoing momenta.



5. *Integration over internal momenta*: For each internal line, write down a factor*

$$\frac{1}{(2\pi)^4} d^4 q_j$$

and integrate over all internal momenta.

6. *Cancel the delta function*: The result will include a delta function

$$(2\pi)^4 \delta^4(p_1 + p_2 + \cdots - p_n)$$

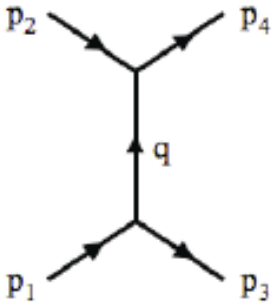
reflecting overall conservation of energy and momentum. Erase this factor† and multiply by i. The result is $\mathcal{M}$.

Summarized

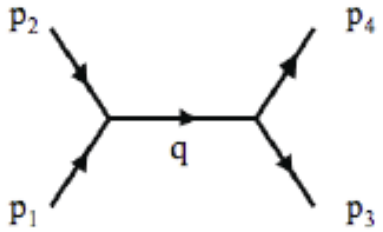
- 1) Diagram – Notate (1,2,3,.etc)
- 2) Each Vertex factor F – coupling g
  → F = -ig
- 3) (Each) propagator $i/(q^2-m^2)$
- 4) Eac vertex ("uvw") $\delta(p_u \pm p_v \pm p_w)$ ie 4v mom conservation 3legs/arrows
- 5) Integrate internal lines d^4q
- 6) Cancel/erase the final delta fn (δ^4) --global 4v mom consv
- 7) Multiply by i

tree diagram δfn's: 2 (8) δfns [two 4v]- 1(4) Integrals-1 (4) final = no "real" integrations!

Toy Theory A+B→A+B



M_1



M_2

$$M_1 \longrightarrow \int (-ig)^2 \left(\frac{i}{q^2 - m_C^2 c^2}\right) (2\pi)^4 \delta^4(p_1 - q - p_3)(2\pi)^4 \delta^4(p_2 + q - p_4) \frac{d^4q}{(2\pi)^4}$$

do integral → apply 2nd δfn to first → sends q = p4-p2

$$\delta \rightarrow \delta^4(p_1 - (p_4 - p_2) - p_3)$$
$$= \delta^4(p_1 + p_2 - p_3 - p_4)$$

Continuing..M_1 A+B→A+B

$$\frac{-ig^2}{(p_1-p_3)^2-m_C^2c^2}(2\pi)^4\delta^4(p_1+p_2-p_3-p_4) = -i\mathcal{M}_1$$

Very common way to denote rule 7 – many text derive M starting with this expression due to other reasons

$$\mathcal{M}_1 = \frac{g^2}{(p_1-p_3)^2-m_C^2c^2}$$

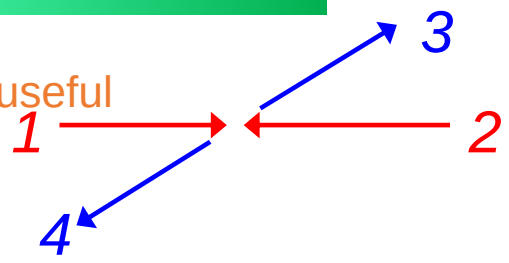
Done with M_1

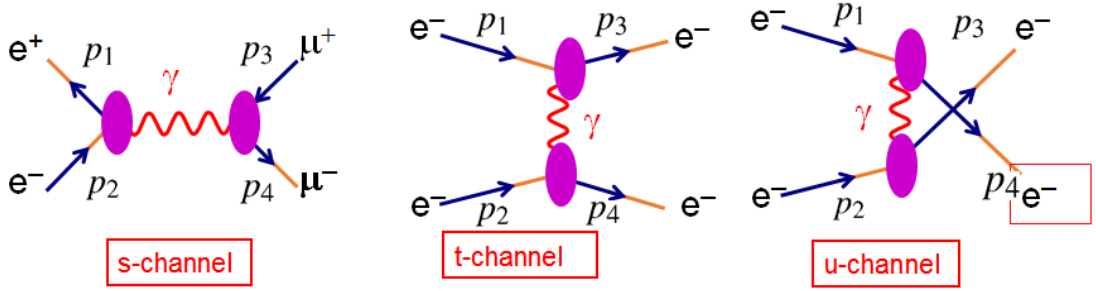
Mendelstaam t !

Review/New : Mandelstam s, t and u

★ In particle scattering/annihilation there are three particularly useful Lorentz Invariant quantities: s, t and u

★ Consider the scattering process $1 + 2 \rightarrow 3 + 4$





$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2$$

Note also

$$s = (p_3 + p_4)^2 \qquad t = (p_2 - p_4)^2 \qquad u = (p_2 - p_3)^2$$

"mom lost = gained"

A few more relations are on the way (besides $s+t+u = \Sigma m_i^2$)

M_2 for A+B→A+B

- (Common move also:) BUT we don't need to do everything again.
- Almost nothing would be different except delta fn's and what q is

$M_2 \rightarrow$ $$\int (-ig)^2 \left(\frac{i}{q^2 - m_C^2 c^2}\right) (2\pi)^4 \delta^4(p_1 - q - p_3)(2\pi)^4 \delta^4(p_2 + q - p_4) \frac{d^4q}{(2\pi)^4}$$

$\delta(p_1 + p_2 - q)$ $\delta(q - p_3 - p_4)$

1

$q = (p_1 + p_2)$

$$\mathcal{M}_2 = \frac{g^2}{(p_1 + p_2)^2 - m_C^2 c^2}$$

Mendelstaam s

Squaring

- Total M has two terms, there is an interference term when modulus taken
- Note these M's are real (not complex)...this will not always be the case

$$|M|^2 = (M_1 + M_2)(M_1^* + M_2^*) = |M_1|^2 + |M_2|^2 + \underbrace{2\,\mathrm{Re}[M_1 M_2^*]}_{\text{Interference}}$$

$$\frac{M}{g^2} = \frac{1}{t - m_c^2} + \frac{1}{s - m_c^2}$$ (both real)

- For m = 0 (like a photon) **OR at high energy**

$$|M|^2 = \frac{g^2}{t^2} + \frac{g^2}{s^2} + \frac{2g^2}{ts}$$

Invariant ! (by construction)

Putting Into Cross Section

- Which x.s formula? Well we must choose frame... however our for ~all formula still need COM momentum calc'd (used in all expressions :)

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathscr{M}|^2}{(E_1+E_2)^2}\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

$(\sqrt{s})^2$

$$\sigma = \frac{1}{64\pi^2 s}\frac{|\vec{p}_f^{\,*}|}{|\vec{p}_i^{\,*}|}\int |M_{fi}|^2 d\Omega^*$$

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s|\vec{p}_i^{\,*}|^2}|M_{fi}|^2$$

In COM frame AND with mA = mB (like last class) << s

$$p_i = \sqrt{\frac{s}{4} - m_B^2}$$

$$\frac{d\sigma}{dt} = \frac{g^2}{32\pi s\sqrt{s-4m^2}}\left(\frac{1}{t^2}+\frac{1}{s^2}+\frac{2}{st}\right)$$

- For dΩ formulae pf = pi for mA= mB

$$\frac{d\sigma}{d\Omega^*} = \frac{1}{64\pi^2 s}\left(\frac{1}{t^2}+\frac{1}{s^2}+\frac{2}{st}\right)$$

Reminder to get angular dep from t there are examples in last weeks slides from the Thomson slides

A+A->B+B

- See Griff for the other main example
- Interesting to point out some features on that one because of crossing

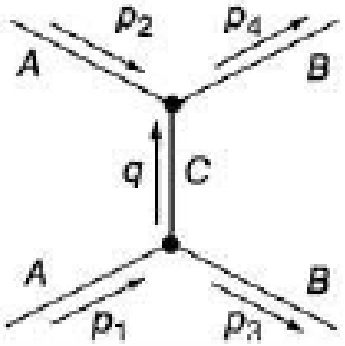


Fig. 6.8 Lowest-order contribution

1 to $A + A \rightarrow B + B$.

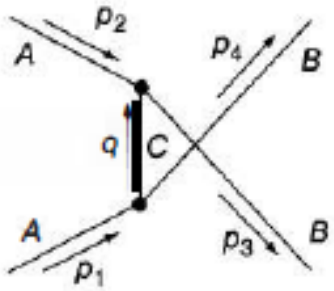
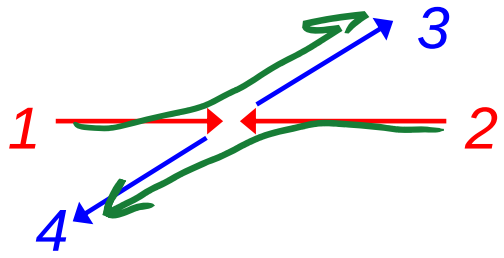
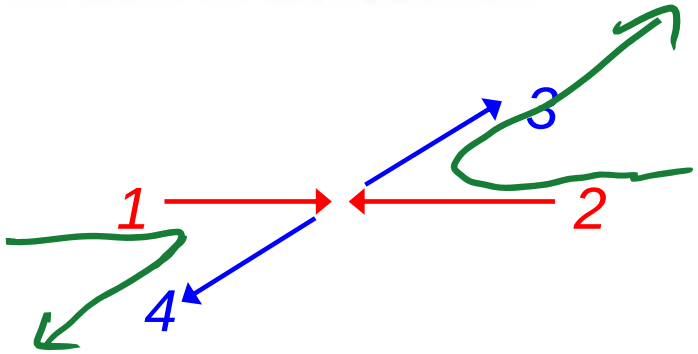


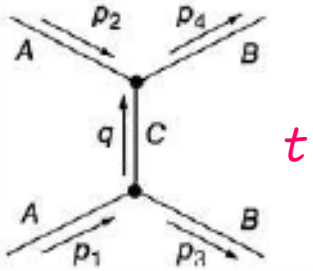
Fig. 6.9 Second diagram contributing in lowest order to $A + A \rightarrow B + B$.





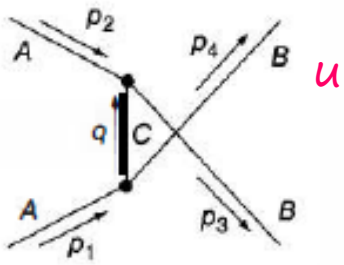
A+A->B+B - Crossing diagrams

- See Griff for this other main example AA→BB
- Interesting to point out some features on that one because of crossing



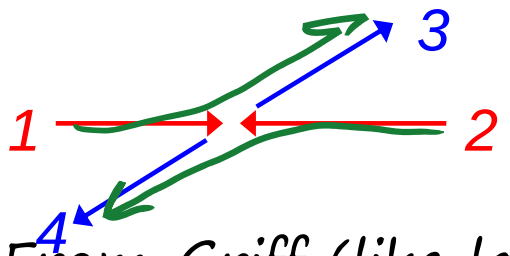
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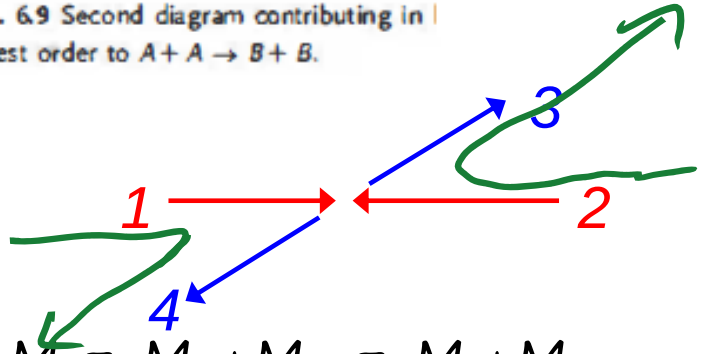
Fig. 6.8 Lowest-order contributior
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u

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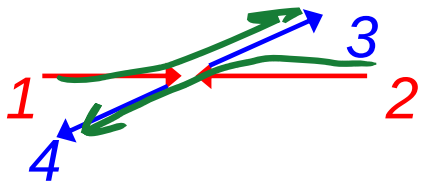
- From Griff (like last example): $M = M_1 + M_2 = M_t + M_u$ (also no "real" integration – only δ's)

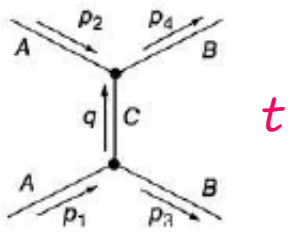
$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_C^2 c^2}$$

$$= g^2 * \frac{1}{t - m^2} + \frac{1}{u - m_c^2}$$

crossing t/u

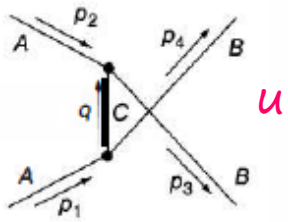
- Now we see a concrete case





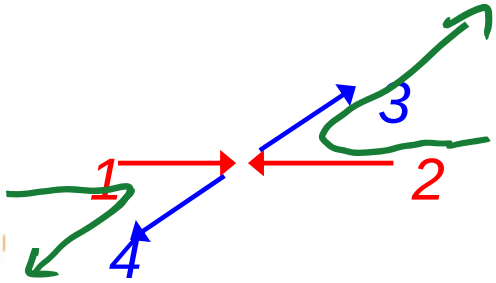
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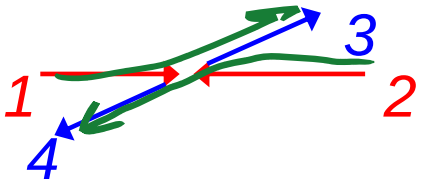
$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_C^2 c^2}$$

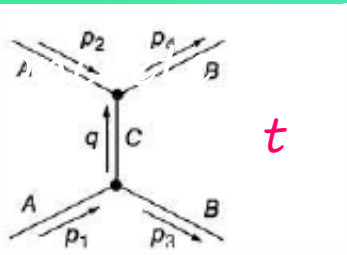
$$= g^2 * \frac{1}{t - m^2} + \frac{1}{u - m_c^2}$$

- Choose again mA=mb; mc = 0 as in Griff ($\propto 1/t+1/u$)
- Recall Pset again $t/u = 2p(1 \pm \cos\theta)$ →

- **Point 2) :** for every angle there are both t and u contributing (QM'ally cannot distinguish) then there is even an interference
  - $M^2 \propto 1/u^2+1/t^2$ +2/tu see 2 slides from now, e.g. θ=0, 1/t diverges,1/u finite at $\theta = 180$ 1/u diverges 1/t finite

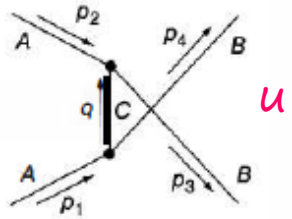
Final calc





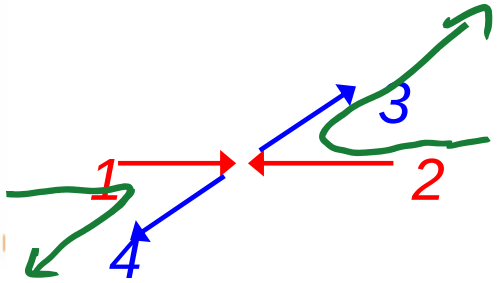
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Fig. 6.8 Lowest-order contributi
1 to A + A → B + B.



u

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$$\mathcal{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_C^2 c^2}$$

$$= g^2 * \frac{1}{t - m^2} + \frac{1}{u - m_c^2}$$

- Comment on t/u coming.... first calc
- Choose again $m_A = m_B$; $m_c = 0$ as in Griff (now $M \propto 1/t + 1/u$)
- Recall Pset again $t/u = 2p(1 \pm \cos\theta)$ →

$$\mathcal{M} = -\frac{g^2}{\mathbf{p}^2 \sin^2\theta} \qquad \frac{d\sigma}{d\Omega} = \frac{1}{2}\left(\frac{\hbar c g^2}{16\pi E \mathbf{p}^2 \sin^2\theta}\right)^2$$

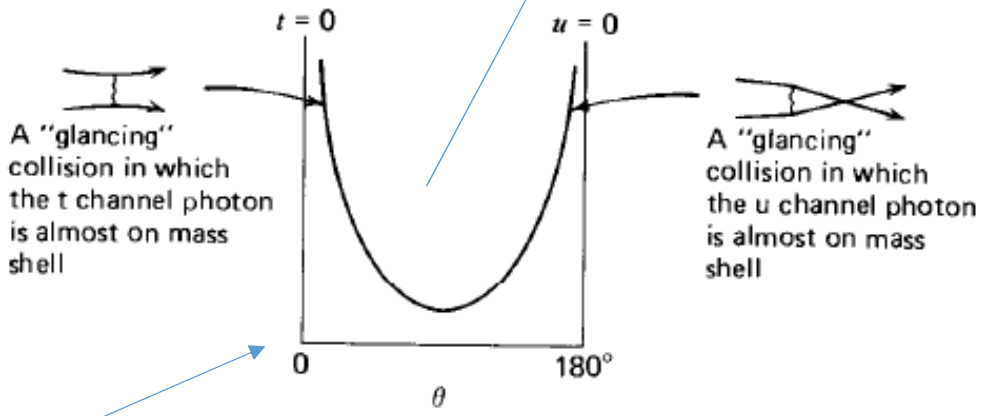
- Very Similar form as Rutherford Scattering (Grad Lab)
- OBVIOUSLY Mandelstam forms are convenient but for usefulness they generally need converted into ANGLE formula
  - Having a proficiency in both is good (angle ← → s,t,u)

Most QED/QCD "FORWARD" PEAKED

- **Point 3)** Note that the infinity may not be "physical" - Griff's comments earlier (EM infinite range)
  - There must be some limit, practically
- **Can even be Divergent (infrared small q^2)**

$$\mathcal{M} = -\frac{g^2}{p^2 \sin^2\theta} \qquad \frac{d\sigma}{d\Omega} = \frac{1}{2}\left(\frac{\hbar c g^2}{16\pi E p^2 \sin^2\theta}\right)^2 \propto \left(\frac{1}{t} + \frac{1}{u}\right)$$

$$\frac{d\sigma}{d\Omega^*} = \frac{1}{64\pi^2 s}\left(\frac{1}{t^2} + \frac{1}{s^2} + \frac{2}{st}\right)$$



This side is forward direction

Fig. 4.8 The differential cross section, $d\sigma/d\Omega$, for electron–electron scattering.

Halzen and Martin

A Couple Lead Ups To TRUE QED

- Scalar QED

Note: Feynman Rules-Shmules?

- **Chp 6 TOY:** We've already done the δ fn counting for tree level (ie 2→2 no loops) diagrams btw 2→ 3 is NLO → UV divergence
  - **No real integral methods needed!!!** NLO/loops are beyond the scope of our course (ability to calc/renorm not too useful e.g. to practical experiementalists—(maybe a few exceptions like 2→3) –but generally a very theorist-specialized skill)
- The complex expressions used for **M** with all the rules, especially with the INTEGRALS are really only needed for ≥ NLO/loops
  - **Allows us to see need for RENORMILIZATION NICELY IN Griff**
- Many other courses (e.g. Halzen and Martin) start with simpler motiviation for tree level calcs from Perturb. Theory (just as I have reminded you of last time)
  - **in a way that involves NO INTEGRATION !!!**
- This is in the context of **"SPINLESS" QED** which is a very nice thing to study as well – so we will, a little too
  - Good intuition before Dirac complications for many phenomena
  - Not

Matrix Element again - focus

- Matrix element so much easier to make (actual (non)"Feynman" rules from another course using Halzen and Martin)

To evaluate the matrix element $\mathcal{M}$ write down

1. the current associated with each of the fermion lines,
2. the propagators associated with each of the bosons,
3. the coupling strengths at each of the interaction vertices.

$\mathcal{M}$ is the product of these terms. Additionally you may have to calculate the four-momentum transfer q and impose four momentum conservation to evaluate the matrix element.

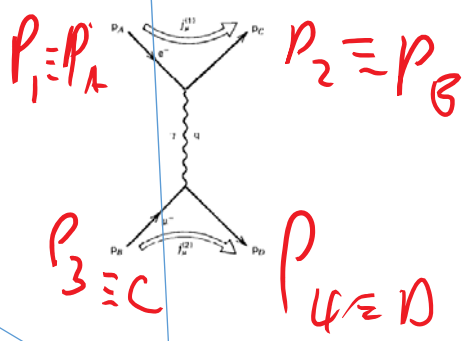
Matrix element:

$jA^{\mu} = j1\ (1/q)\ j2$

Ignore why part of this slide for now just previous easy M tricks/Feynman rules schmules logic

$$-i\mathcal{M} = \left(ie(p_A + p_C)^{\mu}\right)\left(-i\frac{g_{\mu\nu}}{q^2}\right)\left(ie(p_B + p_D)^{\nu}\right).$$

This is regular matrix element!!! for t-channel
$q = t$

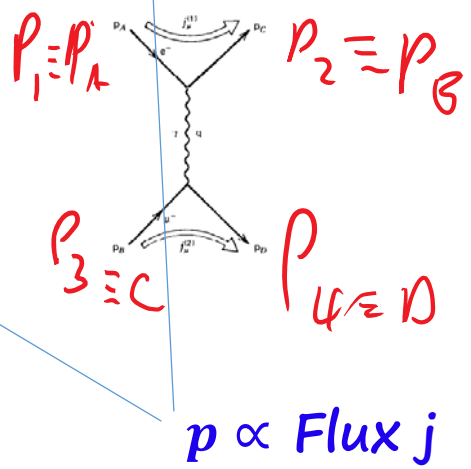


$p \propto$ Flux j

Exercise

$$-i\mathcal{M} = \left(ie(p_A + p_C)^\mu\right)\left(-i\frac{g_{\mu\nu}}{q^2}\right)\left(ie(p_B + p_D)^\nu\right).$$

This is regular matrix element!!! for t-channel
$q = t$



$p \propto$ Flux j

- show in High Energy limit s, t, u are interms of dot products of p1, p2,p3,p4
- Evaluate these products and write what the matrix element is proportional to in terms of s,t,u

Matrix Element again - focus

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Matrix element:

$jA^{\mu} = j1\ (1/q)\ j2$

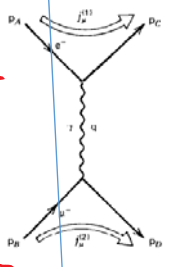
Ignore why part of this slide for now just previous easy M tricks/Feynman rules schmules logic

$$-i\mathcal{M} = \left(ie(p_A + p_C)^{\mu}\right)\left(-i\frac{g_{\mu\nu}}{q^2}\right)\left(ie(p_B + p_D)^{\nu}\right).$$

This is regular matrix element!!! for t-channel
$q = t$

t

$p_1 = p_A$ $p_2 = p_B$ $p_3 = C$ $p_4 = D$



$p_1 \cdot p_2 + p_1 \cdot p_4 + p_3 \cdot p_2 + p_3 \cdot p_4 / t$

$s - u$

$M \propto \frac{s-u}{t}$

$p \propto$ Flux j

Dirac → Jibberish→ j

- Considering that $u(p)/\psi$ are already mixture of spin and p then we need to make these extra sandwiches seems like these are jibberish!
- In fact I will call the kind of terms jibberish terms sometimes

$$-e\bar{\psi}_f \gamma_\mu \psi_i$$

$$-e\bar{u}_f \gamma_\mu u_i e^{i(p_f - p_i)\cdot x}$$

Flux j

but we will find there not as jibberish-y as they look

Exercise

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

$\sim -1\mathbb{I}_{2\times 2}$

- For Dirac

$$j^\mu = \bar{\psi}\gamma^\mu\psi$$

- It turns out that each term in this when using ANY of the wave fn's u/v

$$j^\mu = \bar{u}_i\gamma^\mu u_i = 2p^\mu$$

- where i is a constant integer < 4, ie each component of the RHS is a cyclic permutation of the component on the LHS.
- Using any of the wave fn solutions (choose one) show that this is true for one of the components also of your choice and answer in tophat the value of i based on that.

$$u_1 = N\begin{pmatrix} 1 \\ 0 \\ \dfrac{c(p_z)}{E+mc^2} \\ \dfrac{c(p_x+ip_y)}{E+mc^2} \end{pmatrix} \qquad u_2 = N\begin{pmatrix} 0 \\ 1 \\ \dfrac{c(p_x-ip_y)}{E+mc^2} \\ \dfrac{c(-p_z)}{E+mc^2} \end{pmatrix}$$

$$v^{(1)} = N\begin{pmatrix} \dfrac{c(p_x-ip_y)}{E+mc^2} \\ \dfrac{c(-p_z)}{E+mc^2} \\ 0 \\ 1 \end{pmatrix}, \qquad v^{(2)} = -N\begin{pmatrix} \dfrac{c(p_z)}{E+mc^2} \\ \dfrac{c(p_x+ip_y)}{E+mc^2} \\ 1 \\ 0 \end{pmatrix}$$

$$\psi = ae^{-ip\cdot x/\hbar}u \quad \text{(particles)}, \qquad \psi = ae^{ip\cdot x/\hbar}v \quad \text{(anti}$$

N=sqrt(E+m)

it is possible also to show this without the expclicit forms feel free to try that way...too to check?

Map of Dirac Equation Complications

- γ Matrices
- Spinors (4D)
- Plain (plane) Wave Spinor Solutions u and v
- "Bilinear Covariants"

- Feynman rules – ~~Photon Spin!!!!~~ (not important for now, propagator today just $\propto 1/q^2$)

- Trace Theorems- Casimir's Trick

Factor full 4v

- For full four vectors enter vector "factors" β ,γ

$$\left(p^{\mu}p_{\mu} - m^2c^2\right) = \left(\beta^{\kappa}p_{\kappa} + mc\right)\left(\gamma^{\lambda}p_{\lambda} - mc\right)$$

$(= 0)$

$$\beta^{\kappa}\gamma^{\lambda}p_{\kappa}p_{\lambda} - m^2c^2 - mc\left(\beta^{\kappa}p_{\kappa} - \gamma^{\lambda}p_{\lambda}\right)$$

0 if $\beta^{\kappa} = \gamma^{\lambda}$

- Then we want: $\gamma^{\kappa}\gamma^{\lambda}p_{\kappa}p_{\lambda} = p^{\mu}p_{\mu}$
- This will happen if $\gamma^{\mu}\gamma^{\nu} = g^{\mu\nu}$
- or if [since p_k's are just numbers they can be interchanged and then

$$\gamma^{\kappa}\gamma^{\lambda}p_{\kappa}p_{\lambda} = \frac{1}{2}\left(\gamma^{\kappa}\gamma^{\lambda}p_{\kappa}p_{\lambda} + \gamma^{\kappa}\gamma^{\lambda}p_{\lambda}p_{\kappa}\right) = \frac{1}{2}\left\{\gamma^{\kappa},\gamma^{\lambda}\right\}p_{\kappa}p_{\lambda}$$

now rewrite $\gamma^{\lambda}\gamma^{\kappa}p_{\kappa}p_{\lambda}$

- so...] or if

$$\left\{\gamma^{\mu},\gamma^{\nu}\right\} = 2g^{\mu\nu}$$

- A key defining property of these gamma matrices

Gamma Matrices

- 2x2 block form

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \qquad \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

- Note other convention mentioned in HW
- TRACELESS ! (Traces will be important later)
- Trace of any ODD number of γ's is also 0 Example next slide
- In Dirac Equation looks like the following:

Expand this and then write in 2x2 matrix form: $\gamma^\mu p_\mu = \gamma^0 p_0 - \vec{\gamma}\cdot\vec{p}$

$$= \frac{E}{c}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \vec{p}\cdot\begin{pmatrix} 0 & \vec{\sigma} \\ -\vec{\sigma} & 0 \end{pmatrix} = \begin{pmatrix} E/c & -\vec{p}\cdot\vec{\sigma} \\ \vec{p}\cdot\vec{\sigma} & -E/c \end{pmatrix} \qquad \left[\gamma^i = \begin{pmatrix} 0 & \vec{\sigma}_i \\ -\vec{\sigma}_i & 0 \end{pmatrix}\right.$$

One Additional For Parity

- One further gamma matrix will play a big role

$$\bar{\psi}\gamma^5\psi \text{ with } \gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$$

$$\gamma^5 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$ in the usual 2x2 form

- Alt convention (Weyl) for gamma

= (-1) x (our γ^0)

$$\gamma^0 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \Rightarrow \quad \gamma^5 = \begin{pmatrix} -1_2 & \\ & 1_2 \end{pmatrix}.$$

- Other Related Operators

Spin (Operator)

$$\vec{S} = \frac{1}{2}\vec{\Sigma} = \frac{1}{2}\begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}$$

$\hat{C} = i\gamma^2$ Charge Conj

$\hat{P} = \gamma^0$ Parity !

The Proof I couldn't Complete

- Trace of all odd numbered products of γ = 0
  - OK I'll just prove it for 1 gamma (already proven γ's traceless)
  - ...and for 3 gammas
- It actually uses BOTH parts of the anti-commutation results of gamma's (meaning:)

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

$$\Rightarrow \begin{cases} \{\gamma^\mu, \gamma^\nu\} = 0 & \text{if } \mu \neq \nu \\ \pm 2 & \text{if } \mu = \nu = 0 \end{cases}$$

i.e. $\gamma^\mu\gamma^\nu = -\gamma^\nu\gamma^\mu$

$\{\gamma^0, \gamma^0\} = 2g^{00}$

$2(\gamma^0)^2 = 2$

$(\gamma^0)^2 = 1$

- Also traces are cyclical meaning Tr(ABC) = Tr(CAB) ... (any number of matrices too)
- PROOF:

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\alpha) = \mathrm{Tr}(\gamma^0\gamma^0\gamma^\mu\gamma^\nu\gamma^\alpha) =$$

$$= \mathrm{Tr}(\gamma^0\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^0) = -\mathrm{Tr}(\gamma^0\gamma^\mu\gamma^\nu\gamma^0\gamma^\alpha)$$

cyclic

commute 2 1

$+1 = (-1)(-1)$

End of Proof

$$= -Tr(\gamma^0 \gamma^0 \gamma^\mu \gamma^\nu \gamma^\alpha) = -Tr(\gamma^\mu \gamma^\nu \gamma^\alpha)$$

$\gamma^0 \gamma^0 = \mathbb{1}$ (again)

- because the end expression is the negative of the starting expression and they are equal the expression must be equal to zero.

- It's a weird proof, also relies seemingly on use of specifically $\gamma 0$ (but certainly γ^5 can be used as well b.c. γ^5^2 also =1, and probably even γ^i even though for 1,23, (γ^i)^2 = -1 – the extra minus sign just gets reabsorbed at the end) but crux is any odd number of (anti) commutes will produce the net negative sign.
  - So this way it's also proven for all odd number of gamma products

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- Trace Theorems- Casimir's Trick

4x4 → 4 Spinors

Now we have $p^\mu p_\mu - m^2c^2 = \left(\gamma^\kappa p_\kappa + mc\right)\left(\gamma^\lambda p_\lambda - mc\right) = 0$ as desired.

Can choose either factor to define the Dirac equation.

By convention $\boxed{\gamma^\mu p_\mu - mc = 0}$.

$$\gamma^\mu \partial_\mu \psi + im\psi = 0$$

Making the usual operator substitution $p_\mu \to i\hbar\partial_\mu$ and allowing the resulting expression to operate on the (now necessarily four component) wavefunction ψ yields the Dirac equation:

$$i\hbar\gamma^\mu\partial_\mu\psi - mc\psi = 0 \qquad \psi = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix}$$

Free particle solution

It is important to note that while the Dirac spinor ψ is a four component object, is it NOT a four-vector. It's transformation properties are important

(it has more like 8 components actually because it's complex!)

could probably γ's Ψ's with real-only but it would be ~8 dimensional?

Plane Wave Spinor Solutions (Summary)

- Plane Wave fn

$$\psi(\vec{r},t) = ae^{-\frac{i}{\hbar}(Et-\vec{p}\cdot\vec{r})}u(E,\vec{p})$$

$u(E,\vec{p})$: 4D spinor part

$ae^{-\frac{i}{\hbar}(Et-\vec{p}\cdot\vec{r})}$: normalization factor: ignore for now

in Dirac eq:

$$\gamma^\mu p_\mu a e^{-\frac{i}{\hbar}(x\cdot p)}u - mcae^{-\frac{i}{\hbar}(x\cdot p)}u = 0 \qquad \boxed{\left(\gamma^\mu p_\mu - mc\right)u = 0}.$$

SOLUTIONS: (Griff ≈ ~most commonly used—not only ones)

$$u_1 = N\begin{pmatrix}1\\0\\\dfrac{c(p_z)}{E+mc^2}\\\dfrac{c(p_x+ip_y)}{E+mc^2}\end{pmatrix} \qquad u_2 = N\begin{pmatrix}0\\1\\\dfrac{c(p_x-ip_y)}{E+mc^2}\\\dfrac{c(-p_z)}{E+mc^2}\end{pmatrix}$$

$$E^2 = +\sqrt{m^2c^4 + |\vec{p}|^2c^2}$$

$$v^{(1)} = N\begin{pmatrix}\dfrac{c(p_x-ip_y)}{E+mc^2}\\\dfrac{c(-p_z)}{E+mc^2}\\0\\1\end{pmatrix}, \qquad v^{(2)} = -N\begin{pmatrix}\dfrac{c(p_z)}{E+mc^2}\\\dfrac{c(p_x+ip_y)}{E+mc^2}\\1\\0\end{pmatrix}$$

$\psi = ae^{-ip\cdot x/\hbar}u$ (particles), $\psi = ae^{ip\cdot x/\hbar}v$ (antiparticle

NORM

$$N \equiv \sqrt{(E+mc^2)/c} \rightarrow u^\dagger u = 2|E|/c$$

Alt Chiral/Weyl Rep: u/v

- Form of u(p) used in L/B and Peskin ("the Chiral/Weyl" choice that seems to accompany the alt choice of γ^μ matrices)

$$u^s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma}\xi^s \\ \sqrt{p \cdot \bar{\sigma}}\xi^s \end{pmatrix}$$

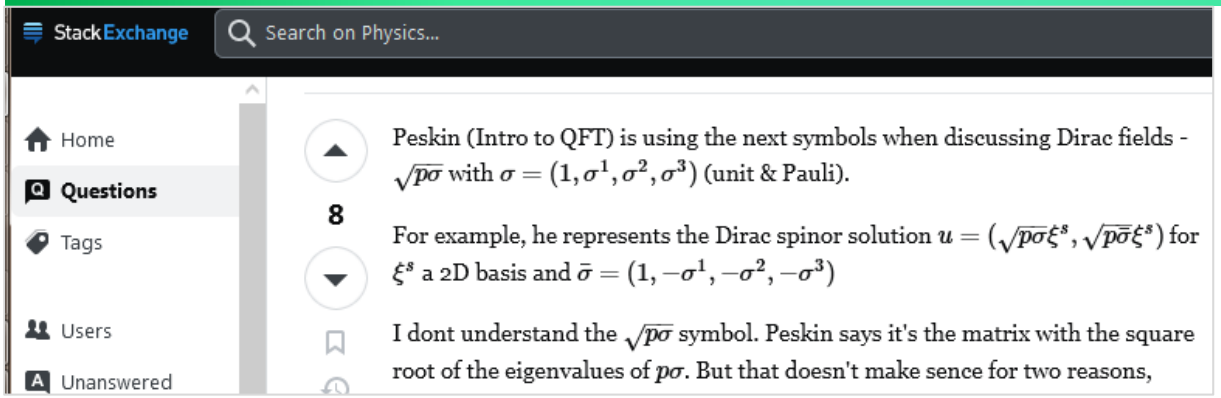
$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

- What does this mean? Annoyingly, never explained in either book [surprising for L/B] , and no explicit form is ever used. Apparently one can get by with only using $u^\dagger u$ in most cases needed in those books. (why p-σ order instead of σ-p?)
- Here is an expression found on Stack Exchange

$$\sqrt{p.\sigma} \equiv \frac{E_p \pm m - \sigma.\mathbf{p}}{\sqrt{2(E_p + m)}}$$

- Different, more symmetric choice than Griff/More standard choice we are using

Preferred u(p)'s for



5

Take a self-adjoint matrix A. There exists a unitary matrix U such that UAU^* is diagonal. Take the square root of every diagonal element in order to define $\sqrt{UAU^*}$ (now you are allowing for roots of negative numbers, so imaginary numbers as well). Then *rotate* the matrix back with U^* and set

$$\sqrt{A} := U^* \sqrt{UAU^*} U.$$

3

The form given in Peskin and Schroeder is useless except in the high energy and low energy limits and to remind the reader that taking the square of these matrices gives the result you would expect (e.g., $\sqrt{p.\sigma}^2 = p.\sigma$). In practice a more useful form to use is,

$$\sqrt{p.\sigma} \equiv \frac{E_p \pm m - \sigma.\mathbf{p}}{\sqrt{2(E_p + m)}}$$

These can be derived from the other answer (though this is not a trivial exercise).

Plane Wave Spinor Solutions

- Plane Wave fn

$$\psi(\vec{r},t) = ae^{-\frac{i}{\hbar}(Et-\vec{p}\cdot\vec{r})}u(E,\vec{p})$$

normalization factor: ignore for now

4D spinor part

in Dirac eq:

$$\gamma^\mu p_\mu a e^{-\frac{i}{\hbar}(x\cdot p)}u - mcae^{-\frac{i}{\hbar}(x\cdot p)}u = 0$$

$$\boxed{\left(\gamma^\mu p_\mu - mc\right)u = 0}.$$

SOLUTIONS: (Griff ≈ ~most commonly used—not only ones)

$$u_1 = N\begin{pmatrix}1\\0\\\dfrac{c(p_z)}{E+mc^2}\\\dfrac{c(p_x+ip_y)}{E+mc^2}\end{pmatrix} \qquad u_2 = N\begin{pmatrix}0\\1\\\dfrac{c(p_x-ip_y)}{E+mc^2}\\\dfrac{c(-p_z)}{E+mc^2}\end{pmatrix}$$

$$E^2 = +\sqrt{m^2c^4 + |\vec{p}|^2 c^2}$$

$$v^{(1)} = N\begin{pmatrix}\dfrac{c(p_x-ip_y)}{E+mc^2}\\\dfrac{c(-p_z)}{E+mc^2}\\0\\1\end{pmatrix}, \qquad v^{(2)} = -N\begin{pmatrix}\dfrac{c(p_z)}{E+mc^2}\\\dfrac{c(p_x+ip_y)}{E+mc^2}\\1\\0\end{pmatrix}$$

$\psi = ae^{-ip\cdot x/\hbar}u$ (particles), $\psi = ae^{ip\cdot x/\hbar}v$ (antiparticle

NORM

$$N \equiv \sqrt{(E+mc^2)/c} \rightarrow u^\dagger u = 2|E|/c$$

Follow Other Slides

- For some points about derivation of explicit forms

Start w/ Mom p = 0

- slides from Kreiger Toronto

Solutions to the Dirac Equation

We had $\left(i\hbar\gamma^{\mu}\partial_{\mu} - mc\right)\psi = 0$:

here the γ matrices are 4x4 and ψ is a 4-component Dirac spinor $\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$

Still need to discuss the form of the spinor ψ:

Consider first the case in which ψ is stationary (independent of position)

$$\frac{\partial\psi}{\partial x} = \frac{\partial\psi}{\partial y} = \frac{\partial\psi}{\partial z} = 0$$ e.g. a state with $\vec{p} = 0$ since $p_{\mu} \to i\hbar\partial_{\mu} \equiv i\hbar\left(\frac{1}{c}\frac{\partial}{\partial t}, -\vec{\nabla}\right)$

In this case, the Dirac equation reduces to $\frac{i\hbar}{c}\gamma^{0}\frac{\partial\psi}{\partial t} - mc\psi = 0$ which we can write as:

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \partial\psi_A/\partial t \\ \partial\psi_B/\partial t \end{pmatrix} = -i\frac{mc^2}{\hbar}\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} \quad \text{with} \quad \psi_A = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \quad \psi_B = \begin{pmatrix} \psi_3 \\ \psi_4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \partial\psi_A/\partial t \\ \partial\psi_B/\partial t \end{pmatrix} = -i\frac{mc^2}{\hbar}\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} \quad \text{with} \quad \psi_A = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \quad \psi_B = \begin{pmatrix} \psi_3 \\ \psi_4 \end{pmatrix}$$

four components

two components each

$$\frac{\partial\psi_A}{\partial t} = -i\frac{mc^2}{\hbar}\psi_A \qquad -\frac{\partial\psi_B}{\partial t} = -i\frac{mc^2}{\hbar}\psi_B$$

These equations have straightforward solutions:

$$\psi_A(t) = e^{-i\left(mc^2/\hbar\right)t}\psi_A(0) \qquad \psi_B(t) = e^{+i\left(mc^2/\hbar\right)t}\psi_B(0)$$

Basic quantum mechanics tells us that characteristic time evolution of a quantum state of energy E is given by $e^{-iEt/\hbar}$: Here $E=mc^2$ for the time dependence of ψ_A. However, for ψ_B we have

$$\psi_B(t) = e^{+i\left(mc^2/\hbar\right)t}\psi_B(0)$$

Interpret these “negative energy” solutions as being positive energy solutions for anti-particles.

So, ψ_A describes electrons and ψ_B describes positrons: each has two spin states:

$$\psi_1 = e^{-i\frac{mc^2}{\hbar}t}\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$$ electron with spin up

$$\psi_2 = e^{-i\frac{mc^2}{\hbar}t}\begin{pmatrix}0\\1\\0\\0\end{pmatrix}$$ electron with spin down

$$\psi_3 = e^{+i\frac{mc^2}{\hbar}t}\begin{pmatrix}0\\0\\1\\0\end{pmatrix}$$ positron with spin up

$$\psi_4 = e^{+i\frac{mc^2}{\hbar}t}\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$ positron with spin down

Now, consider general state (non-stationary): plane wave solutions of form

$$\psi\left(\vec{r},t\right) = ae^{-\frac{i}{\hbar}\left(Et-\vec{p}\cdot\vec{r}\right)}u\left(E,\vec{p}\right)$$

↓

normalization factor: ignore for now

$$\psi\left(\vec{r},t\right)=ae^{-\frac{i}{\hbar}\left(Et-\vec{p}\cdot\vec{r}\right)}u\left(E,\vec{p}\right) \quad \Rightarrow \quad \psi\left(x\right)=ae^{-\frac{i}{\hbar}\left(x\cdot p\right)}u\left(p\right)$$ [in 4v notation]

$\partial_\mu\psi=-\frac{i}{\hbar}p_\mu ae^{-\frac{i}{\hbar}\left(x\cdot p\right)}u\left(p\right)$ so the Dirac equation $\left(i\hbar\gamma^\mu\partial_\mu-mc\right)\psi=0$ becomes

$\gamma^\mu p_\mu ae^{-\frac{i}{\hbar}\left(x\cdot p\right)}u-mcae^{-\frac{i}{\hbar}\left(x\cdot p\right)}u=0$ or, more compactly, $\boxed{\left(\gamma^\mu p_\mu-mc\right)u=0}$.

This is referred to as the Dirac equation in momentum space. Note that this is purely algebraic (i.e. it contains no derivatives). If u satisfies this equation then ψ satisfies the Dirac equation.

Expand this and then write in 2x2 matrix form: $\gamma^\mu p_\mu=\gamma^0 p_0-\vec{\gamma}\cdot\vec{p}$

$$=\frac{E}{c}\begin{pmatrix}1&0\\0&-1\end{pmatrix}-\vec{p}\cdot\begin{pmatrix}0&\vec{\sigma}\\-\vec{\sigma}&0\end{pmatrix}=\begin{pmatrix}E/c&-\vec{p}\cdot\vec{\sigma}\\\vec{p}\cdot\vec{\sigma}&-E/c\end{pmatrix} \qquad \left[\gamma^i=\begin{pmatrix}0&\vec{\sigma}_i\\-\vec{\sigma}_i&0\end{pmatrix}\quad i=1,2,3\right]$$

$$\left(\gamma^{\mu} p_{\mu} - mc\right)u = 0 \quad \Rightarrow \quad \begin{pmatrix} \frac{E}{c} - mc & -\vec{p}\cdot\vec{\sigma} \\ \vec{p}\cdot\vec{\sigma} & -\frac{E}{c} - mc \end{pmatrix}\begin{pmatrix} u_A \\ u_B \end{pmatrix} = 0$$

$$\Rightarrow \quad \begin{pmatrix} \left(\frac{E}{c} - mc\right)u_A - \left(\vec{p}\cdot\vec{\sigma}\right)u_B \\ \left(\vec{p}\cdot\vec{\sigma}\right)u_A - \left(\frac{E}{c} + mc\right)u_B \end{pmatrix} = 0$$

e.g. two equations in u_A and u_B.

$$\left(\frac{E}{c} - mc\right)u_A = \left(\vec{p}\cdot\vec{\sigma}\right)u_B \quad \Rightarrow \quad u_A = \frac{c}{E - mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)u_B$$

$$\left(\vec{p}\cdot\vec{\sigma}\right)u_A = \left(\frac{E}{c} + mc\right)u_B \quad \Rightarrow \quad u_B = \frac{c}{E + mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)u_A$$

$$u_A = \frac{c}{E - mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)\frac{c}{E + mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)u_A = \frac{c^2}{E^2 - m^2c^4}\left(\vec{p}\cdot\vec{\sigma}\right)^2 u_A$$

$$u_A = \frac{c}{E - mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)\frac{c}{E + mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)u_A = \frac{c^2}{E^2 - m^2c^4}\left(\vec{p}\cdot\vec{\sigma}\right)^2 u_A$$ Look at factor $\left(\vec{p}\cdot\vec{\sigma}\right)$

$$\vec{p}\cdot\vec{\sigma} = p_x\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + p_y\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + p_z\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{pmatrix}$$

$$\left(\vec{p}\cdot\vec{\sigma}\right)^2 = \begin{pmatrix} p_z^2 + \left(p_x - ip_y\right)\left(p_x + ip_y\right) & p_z\left(p_x - ip_y\right) - p_z\left(p_x + ip_y\right) \\ p_z\left(p_x + ip_y\right) - p_z\left(p_x - ip_y\right) & \left(p_x + ip_y\right)\left(p_x - ip_y\right) + p_z^2 \end{pmatrix}$$

$$= \begin{pmatrix} p_x^2 + p_y^2 + p_z^2 & 0 \\ 0 & p_x^2 + p_y^2 + p_z^2 \end{pmatrix} = \left|\vec{p}\right|^2 \quad \left(e.g.\ \left|\vec{p}\right|^2\ \mathrm{I}_{2x2}\right)$$

$$u_A = \frac{c^2}{E^2 - m^2c^4}\left|\vec{p}\right|^2 u_A \quad \Rightarrow \quad \text{requires } E^2 = m^2c^4 + \left|\vec{p}\right|^2 c^2 \qquad E^2 = \pm\sqrt{m^2c^4 + \left|\vec{p}\right|^2 c^2}$$

We had $u_A = \frac{c}{E - mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)u_B$ $u_B = \frac{c}{E + mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)u_A$ $\vec{p}\cdot\vec{\sigma} = \begin{pmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{pmatrix}$

$$u_A = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow u_B = \frac{c}{E + mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{c}{E + mc^2}\begin{pmatrix} p_z \\ p_x + ip_y \end{pmatrix}$$

$$u_A = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow u_B = \frac{c}{E + mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{c}{E + mc^2}\begin{pmatrix} p_x - ip_y \\ -p_z \end{pmatrix}$$

$$\Bigg\} \quad E^2 = +\sqrt{m^2c^4 + \left|\vec{p}\right|^2 c^2}$$

$$u_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow u_A = \frac{c}{E - mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{c}{E - mc^2}\begin{pmatrix} p_z \\ p_x + ip_y \end{pmatrix}$$

$$u_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \Rightarrow u_A = \frac{c}{E - mc^2}\left(\vec{p}\cdot\vec{\sigma}\right)\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{c}{E - mc^2}\begin{pmatrix} p_x - ip_y \\ -p_z \end{pmatrix}$$

$$\Bigg\} \quad E^2 = -\sqrt{m^2c^4 + \left|\vec{p}\right|^2 c^2}$$

Need energy as stated to avoid blow-up of denominator at $E=mc^2$.

$$u_1 = N\begin{pmatrix} 1 \\ 0 \\ \dfrac{c\left(p_z\right)}{E+mc^2} \\ \dfrac{c\left(p_x+ip_y\right)}{E+mc^2} \end{pmatrix} \qquad u_2 = N\begin{pmatrix} 0 \\ 1 \\ \dfrac{c\left(p_x-ip_y\right)}{E+mc^2} \\ \dfrac{c\left(-p_z\right)}{E+mc^2} \end{pmatrix} \qquad E^2 = +\sqrt{m^2c^4+\left|\vec{p}\right|^2c^2}$$

$$u_3 = N\begin{pmatrix} \dfrac{c\left(p_z\right)}{E-mc^2} \\ \dfrac{c\left(p_x+ip_y\right)}{E-mc^2} \\ 1 \\ 0 \end{pmatrix} \qquad u_4 = N\begin{pmatrix} \dfrac{c\left(p_x-ip_y\right)}{E-mc^2} \\ \dfrac{c\left(-p_z\right)}{E-mc^2} \\ 0 \\ 1 \end{pmatrix} \qquad E^2 = -\sqrt{m^2c^4+\left|\vec{p}\right|^2c^2}$$

Normalization is $u^\dagger u = 2|E|/c$ (recall that we discussed Lorentz invariant wavefunction normalizations a few lectures ago).

$$u_1^\dagger u_1 = N^2 \begin{pmatrix} 1 & 0 & \dfrac{c\left(p_z\right)}{E+mc^2} & \dfrac{c\left(p_x - ip_y\right)}{E+mc^2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \dfrac{c\left(p_z\right)}{E+mc^2} \\ \dfrac{c\left(p_x + ip_y\right)}{E+mc^2} \end{pmatrix} = N^2 \left(1 + \frac{c^2 p_z^2}{\left(E+mc^2\right)^2} + \frac{c^2\left(p_x^2 + p_y^2\right)}{\left(E+mc^2\right)^2} \right)$$

$$N^2 \left(\frac{\left(E+mc^2\right)^2 + c^2\left|\vec{p}\right|^2}{\left(E+mc^2\right)^2} \right) = N^2 \left(\frac{E^2 + 2Emc^2 + m^2c^4 + c^2\left|\vec{p}\right|^2}{\left(E+mc^2\right)^2} \right) = N^2 \left(\frac{2E\left(E+mc^2\right)}{\left(E+mc^2\right)^2} \right) = N^2 \left(\frac{2E}{E+mc^2} \right)$$

(E^2 = $E + mc^2$)^2 + $c^2|\vec{p}|^2$

$u_1^\dagger u_1 = 2\left|E\right| / c$ requires $N = \sqrt{\left(\left|E\right| + mc^2\right) / c}$ (need $|E|$ for u_3, u_4)

Spin Matrices for Dirac Spinors

$$\vec{S} = \frac{\hbar}{2}\vec{\Sigma} \qquad \vec{\Sigma} = \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}$$

u_1 looks like it might represent a spin-up electron, but it does not. u_1 is NOT an eigenvector of S_z, unless we make a specific choice for the direction of the z axis by choosing it along $\hat{p}$.

In this case, $p_x = p_y = 0$ and all four spinors represent states of definite helicity.

For example:

$$u_1 = N\begin{pmatrix} 1 \\ 0 \\ \dfrac{c\left(p_z\right)}{E+mc^2} \\ \dfrac{c\left(p_x + ip_y\right)}{E+mc^2} \end{pmatrix} \Rightarrow N\begin{pmatrix} 1 \\ 0 \\ \dfrac{c\left|\vec{p}\right|}{E+mc^2} \\ 0 \end{pmatrix} \Rightarrow \quad \Sigma_z u_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} N\begin{pmatrix} 1 \\ 0 \\ \dfrac{c\left|\vec{p}\right|}{E+mc^2} \\ 0 \end{pmatrix} = N\begin{pmatrix} 1 \\ 0 \\ \dfrac{c\left|\vec{p}\right|}{E+mc^2} \\ 0 \end{pmatrix}$$

$$\Sigma_z u_1 = N\begin{pmatrix} 1 \\ 0 \\ \dfrac{c|\vec{p}|}{E+mc^2} \\ 0 \end{pmatrix} = \sqrt{\left(|E|+mc^2\right)/c}\begin{pmatrix} 1 \\ 0 \\ \dfrac{c|\vec{p}|}{E+mc^2} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{\left(|E|+mc^2\right)/c} \\ 0 \\ \dfrac{c|\vec{p}|}{E+mc^2}\sqrt{\left(|E|+mc^2\right)/c} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{\left(|E|+mc^2\right)/c} \\ 0 \\ \sqrt{\left(|E|-mc^2\right)/c} \\ 0 \end{pmatrix}$$

since $$\frac{c|\vec{p}|}{E+mc^2}\sqrt{\left(|E|+mc^2\right)/c} = \frac{\sqrt{c^2|\vec{p}|^2}\sqrt{\left(|E|+mc^2\right)/c}}{E+mc^2} = \frac{\sqrt{\left(E^2-m^2c^4\right)}\sqrt{\left(|E|+mc^2\right)/c}}{E+mc^2}$$

$$= \frac{\sqrt{\left(E-mc^2\right)\cancel{\left(E+mc^2\right)}}\sqrt{\cancel{\left(|E|+mc^2\right)}/c}}{\cancel{E+mc^2}} = \sqrt{\left(E-mc^2\right)/c}$$

u_1 and u_3 are spin-up

u_2 and u_4 are spin-down

To represent positron solutions in terms of the physical energy and momentum of the positron, we flip the signs of E and $\vec{p}$ for u_3 and u_4:

$$\psi\left(\vec{r},t\right) = ae^{-\frac{i}{\hbar}\left(Et-\vec{p}\cdot\vec{r}\right)}u\left(-E,-\vec{p}\right)$$

$$u_4 = N\begin{pmatrix} \dfrac{c\left(p_x - ip_y\right)}{E - mc^2} \\ \dfrac{c\left(-p_z\right)}{E - mc^2} \\ 0 \\ 1 \end{pmatrix} \quad \Rightarrow \quad u_4\left(-E,-\vec{p}\right) = N\begin{pmatrix} \dfrac{c\left(-p_x + ip_y\right)}{-E - mc^2} \\ \dfrac{c\left(+p_z\right)}{-E - mc^2} \\ 0 \\ 1 \end{pmatrix} = N\begin{pmatrix} \dfrac{c\left(p_x - ip_y\right)}{E + mc^2} \\ \dfrac{c\left(-p_z\right)}{E + mc^2} \\ 0 \\ 1 \end{pmatrix}$$

The convention is to write positron states (spinors) as v and to forget about u_3 and u_4;

$$v_1\left(E,\vec{p}\right)=u_4\left(-E,-\vec{p}\right)=N\begin{pmatrix}\dfrac{c\left(p_x-ip_y\right)}{E+mc^2}\\ \dfrac{c\left(-p_z\right)}{E+mc^2}\\ 0\\ 1\end{pmatrix}$$

$$v_2\left(E,\vec{p}\right)=\ u_3\left(-E,-\vec{p}\right)=-N\begin{pmatrix}\dfrac{c\left(p_z\right)}{E+mc^2}\\ \dfrac{c\left(p_x+ip_y\right)}{E+mc^2}\\ 1\\ 0\end{pmatrix}$$

where in each case the energy is $E=+\sqrt{m^2c^4+\left|\vec{p}\right|^2c^2}$

u solutions (representing electrons) satisfy $\left(\gamma^\mu p_\mu-mc\right)u=0$

v solutions (representing positrons) satisfy $\left(\gamma^\mu p_\mu+mc\right)v=0$

We will talk more about the properties of these spinors. Most important is that we know what they are, because the Feynman rules we will learn for QED associate a spinor with each external fermion line. Generally we average over spin states, so whether something is spin-up or spin-down is often not so critical.

We will mostly use the spinor in abstract rather than component form, but it is useful to have explicit forms for derivation of their properties.

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Plane Wave Spinor Solutions

- Plane Wave fn

$$\psi(\vec{r},t) = ae^{-\frac{i}{\hbar}(Et-\vec{p}\cdot\vec{r})}u(E,\vec{p})$$

a: normalization factor: ignore for now

$u(E,\vec{p})$: 4D spinor part

in Dirac eq:

$$\gamma^\mu p_\mu a e^{-\frac{i}{\hbar}(x\cdot p)}u - mcae^{-\frac{i}{\hbar}(x\cdot p)}u = 0 \quad \rightarrow \quad \boxed{\left(\gamma^\mu p_\mu - mc\right)u = 0}.$$

Expand this and then write in 2x2 matrix form: $\gamma^\mu p_\mu = \gamma^0 p_0 - \vec{\gamma}\cdot\vec{p}$

$$= \frac{E}{c}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \vec{p}\cdot\begin{pmatrix} 0 & \vec{\sigma} \\ -\vec{\sigma} & 0 \end{pmatrix} = \begin{pmatrix} E/c & -\vec{p}\cdot\vec{\sigma} \\ \vec{p}\cdot\vec{\sigma} & -E/c \end{pmatrix} \qquad \left[\gamma^i = \begin{pmatrix} 0 & \vec{\sigma}_i \\ -\vec{\sigma}_i & 0 \end{pmatrix}\right.$$

SOLUTIONS: (Griff ≈ ~most commonly used—not only ones)

$$u_1 = N\begin{pmatrix} 1 \\ 0 \\ \frac{c(p_z)}{E+mc^2} \\ \frac{c(p_x+ip_y)}{E+mc^2} \end{pmatrix} \qquad u_2 = N\begin{pmatrix} 0 \\ 1 \\ \frac{c(p_x-ip_y)}{E+mc^2} \\ \frac{c(-p_z)}{E+mc^2} \end{pmatrix}$$

$$v^{(1)} = N\begin{pmatrix} \frac{c(p_x-ip_y)}{E+mc^2} \\ \frac{c(-p_z)}{E+mc^2} \\ 0 \\ 1 \end{pmatrix}, \qquad v^{(2)} = -N\begin{pmatrix} \frac{c(p_z)}{E+mc^2} \\ \frac{c(p_x+ip_y)}{E+mc^2} \\ 1 \\ 0 \end{pmatrix}$$

$\psi = ae^{-ip\cdot x/\hbar}u$ (particles), $\psi = ae^{ip\cdot x/\hbar}v$ (antiparticles)

$$E^2 = +\sqrt{m^2c^4 + |\vec{p}|^2 c^2}$$

NORM

$$N \equiv \sqrt{(E+mc^2)/c} \rightarrow u^\dagger u = 2|E|/c$$

Map of Dirac Equation Complications

- γ Matrices
- Spinors (4D)
- Plain (plane) Wave Spinor Solutions u and v
- "Bilinear Covariants"

NEXT TIME

- Feynman rules – ~~Photon Spin!!!!~~ (not important for now, propagator today just $\propto 1/q^2$

- Trace Theorems- Casimir's Trick

Bilinear Covariants

- 4-spinors not 4-v's, how to transform (need whole Dirac eq covariant/invariant) $i\hbar\gamma^{\mu}\partial_{\mu}\psi - mc\psi = 0$
- Can't transform $|\psi|^2$ properly though with just a single regular transformation matrix/operator– closest we can come is as follows—which we will eventually use but with a fix

$$\psi \to \psi' = S\psi \qquad S = \begin{pmatrix} a_+ & a_-\sigma_1 \\ a_-\sigma_1 & a_+ \end{pmatrix}$$

Note that this is a 4x4 matrix. The σ_i are the 2x2 Pauli spin matrices (here we use σ_1 for the case of a boost along x).

works for ψ but not for $\psi^{\dagger}\psi$

with $a_{\pm} = \pm\sqrt{(\gamma \pm 1)/2}$ and $\gamma = (1-\beta^2)^{-1/2}$

Btw normally (as w/ 2-D spin) we would create probability "$\rho(x)$" = $|\psi|^2$ like this:

$$\psi^{\dagger}\psi = \begin{pmatrix} \psi_1^* & \psi_2^* & \psi_3^* & \psi_4^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = |\psi_1|^2 + |\psi_2|^2 + |\psi_3|^2 + |\psi_4|^2$$

But this is will not transform with the above matrix

As usual borrowing much from Kreiger Toronto

Matrix Elements of Operators

- Similarly any time we want to do a matrix element of an operator, sandwich between $\bar{\psi}$ and ψ

$$\langle \psi_a | \hat{O} | \psi_b \rangle \equiv \bar{\psi} \hat{O} \psi$$

Example?!? (sort of) Parity, the operator, in this matrix rep, **it IS γ^0!!!**

$$\psi \rightarrow \psi' = \gamma^0 \psi$$

So our $\bar{\psi}\psi$ is an regular scalar (and invariant)

$$(\bar{\psi}\psi) \rightarrow (\bar{\psi}\psi)' = (\psi^\dagger)' \gamma^0 \psi' = \psi^\dagger (\gamma^0)^\dagger \gamma^0 \gamma^0 \psi = \psi^\dagger \gamma^0 \psi = \bar{\psi}\psi$$

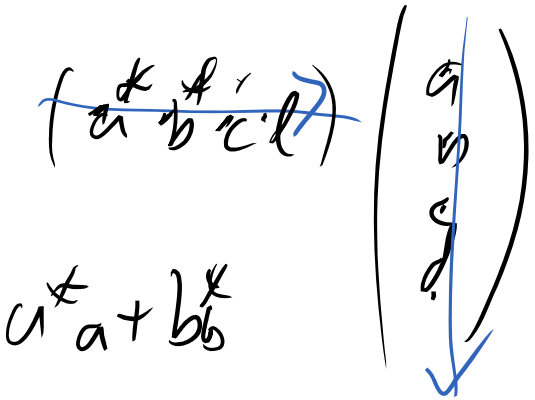
PS ψ is not an eigenvector of parity (and expectation value is not a number or a Lorentz invariant)

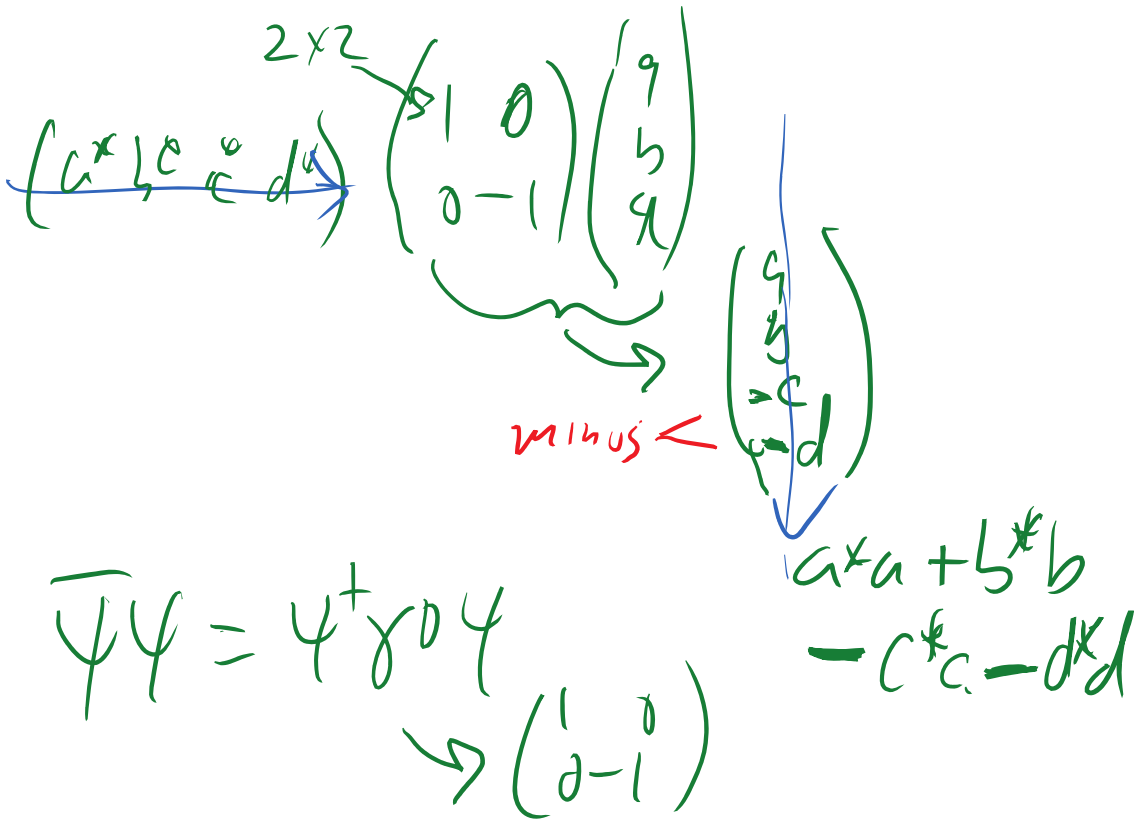
$$\bar{\psi} \gamma^0 \psi = \psi^\dagger \gamma^0 \gamma^0 \psi = \psi^\dagger \psi$$

Mechanically

- compare/view visually $\psi^\dagger\psi \rightarrow \psi^\dagger\gamma^0\gamma^0\psi = \bar{\psi}\psi$

$$(\bar{\psi}\psi) \rightarrow (\bar{\psi}\psi)' = (\psi^\dagger)'\gamma^0\psi' = \psi^\dagger(\gamma^0)^\dagger\gamma^0\gamma^0\psi = \psi^\dagger\gamma^0\psi = \bar{\psi}\psi$$





Parity → Pseudoscalar/pseudovectors

- We can make a pseudoscalar with yet one more gamma matrix

$$\psi \rightarrow \psi' = \gamma^0 \psi$$

Construct other constructs by sandwiching other gammas:

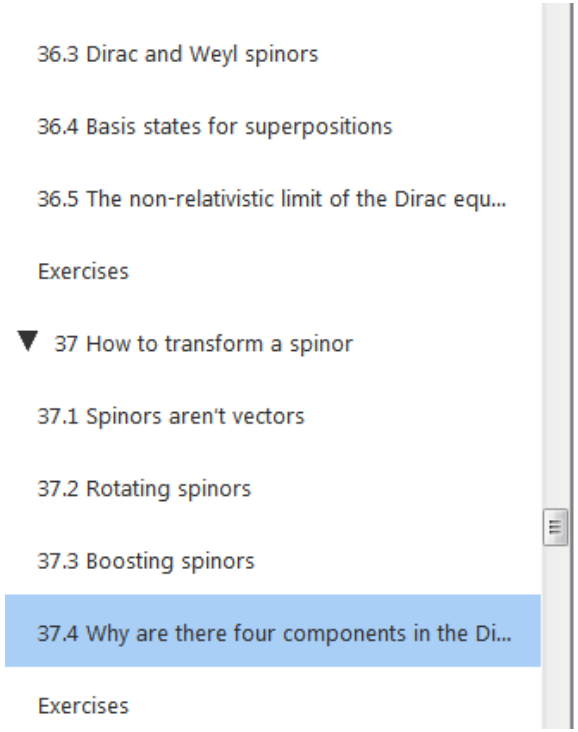
(1)	$\overline{\psi}\psi$ = scalar	(one component)
(2)	$\overline{\psi}\gamma^5\psi$ = pseudoscalar	(one component)
(3)	$\overline{\psi}\gamma^\mu\psi$ = vector	(four components)
(4)	$\overline{\psi}\gamma^\mu\gamma^5\psi$ = pseudovector	(four components)
(5)	$\overline{\psi}\sigma^{\mu\nu}\psi$ = antisymmetric tensor	(six components)

We will be interested in γ^5 later (esp. for Weak int)

Where did Parity come from?

- Why did we suddenly insert parity into the conversation:

- See QFT Lancaster/Blundell chapter:
  - In many limits, the 4-D spinor becomes like a repeat of two 2-spinors e.g. p→∞ (1010) so you may ask :



37.4 Why are there four components in the Dirac equation?

The Dirac equation is written in terms of a four-component Dirac spinor ψ. These Dirac spinors have two degrees of freedom: the two components of ξ (or η). So why do we have four components in the Dirac equation?

We found that for the case of massless particles the Dirac spinors fell apart into two: left-handed Weyl spinors and right-handed Weyl spinors. Each of these spinors carried the same spin information in ξ (or η). The question is, then, why does the world need left- and right-handed fields. We will answer this question by asking what fundamental operation turns left-handed spinors into right-handed spinors? The answer is parity.

Many of the properties of fermions are invariant with respect to parity and so we require that Dirac particles have the same properties in a parity reversed world. (The weak interaction violates parity, but we ignore that for now.) The parity operation P does the following: it maps $x \to -x$ and $v \to -v$ and so it results in the changes to the generators:

$$\mathsf{P}^{-1}\boldsymbol{K}\mathsf{P} = -\boldsymbol{K}, \quad \mathsf{P}^{-1}\boldsymbol{J}\mathsf{P} = +\boldsymbol{J} \qquad (37.20)$$

"the answer is parity ..."

How about vector

- All these examples are related to the probabily density

- What is the vector there?

(1)	$\overline{\psi}\psi$ = scalar	(one component)
(2)	$\overline{\psi}\gamma^5\psi$ = pseudoscalar	(one component)
(3)	$\overline{\psi}\gamma^\mu\psi$ = vector	(four components)
(4)	$\overline{\psi}\gamma^\mu\gamma^5\psi$ = pseudovector	(four components)
(5)	$\overline{\psi}\sigma^{\mu\nu}\psi$ = antisymmetric tensor	(six components)

- It will turn out to be J
- It's just the j^μ !!!! The flux for the case of Dirac particles

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Real Dirac QED : Feyman Rules

- The Feyman rules for QED compared to ABC prescribe the insertion of an extra factor (in addition to ig) for each vertex / incoming/outgoing legs that involves the spinors:

2. *Externallines:* Externallines contribute factors as follows:

-Griff

Electrons: Incoming(⟶•) : u; Outgoing(•⟶) : $\bar{u}$

Positrons: Incoming(⟵•) : $\bar{v}$; Outgoing(•⟵)

3. *Vertex factors:* Each vertex contributes a factor

$ig_e\gamma^\mu$

$$ig_e\gamma^\mu$$

The dimensionless coupling constant g_e is related to the charge of the electron: $g_e = e\sqrt{4\pi/\hbar c} = \sqrt{4\pi\alpha}$.*

4. *Propagators:* Each internal line contributes a factor as follows:

Photons: $\dfrac{-ig_{\mu\nu}}{q^2}$

- when put together these parts have this form :

$$(2\pi)^4 \int [\bar{u}^{(s3)}(p_3)(ig_e\gamma^\mu)u^{(s1)}(p_1)]\frac{-ig_{\mu\nu}}{q^2}[\bar{u}^{(s4)}(p_4)(ig_e\gamma^\nu)u^{(s2)}(p_2)]$$

More

Feynman Rules for QED cont'd

2. External Lines: contribute factors to $\mathcal{M}$ as follows:

• electrons (e.g. fermions)	incoming	u	
	outgoing	$\bar{u} \equiv u^\dagger \gamma^0$	
• positrons (e.g. anti-fermions)	incoming	v	
	outgoing	$\bar{v} \equiv v^\dagger \gamma^0$	
• photons	incoming	ε^μ	polarization vectors: see § 7.4. We will discuss this in an upcoming lecture.
	outgoing	$\varepsilon^{\mu *}$	

3. Vertices: each vertex contributes a factor of $ig\gamma^\mu$ (photon is spin-1).
(here my g is Griffiths $g_e \equiv e\sqrt{4\pi / \hbar c} = \sqrt{4\pi\alpha}$).

4. Propagators: each internal line contributes a factor of:

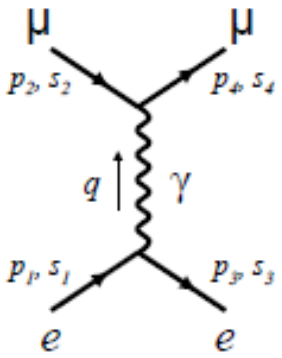
- electrons and positrons $\dfrac{i\left(\gamma^\mu q_\mu + mc\right)}{q^2 - m^2c^2}$ [e.g. internal fermion line]
- photons $-i\dfrac{g_{\mu\nu}}{q^2}$

Vertex ubarXu form

Electron-Muon Scattering in QED

Now back to $e^- \mu^- \to e^- \mu^-$ scattering in QED:

We had:



Now apply Feynman rules to obtain $\mathcal{M}$.

Procedure is to write down terms working backwards in time along each fermion line:

electron line: $\overline{u}^{(s_3)}(p_3)\, ig\gamma^{\mu}\, u^{(s_1)}(p_1)(2\pi)^4 \delta^4(p_1 - p_3 - q)$

outgoing electron spinor	vertex coupling	incoming electron spinor	δ-function for conservation of energy and momentum at electron vertex.

muon line: $\overline{u}^{(s_4)}(p_4)\, ig\gamma^{\nu}\, u^{(s_2)}(p_2)(2\pi)^4 \delta^4(p_2 + q - p_4)$

Mixing of p/spin ? Jibberish?

- Considering that u(p)/ψ are already mixture of spin and p then we need to make these extra sandwiches seems like these are jibberish!
- In fact I will call the kind of terms jibberish terms sometimes

$$-e\bar{\psi}_f \gamma_\mu \psi_i$$

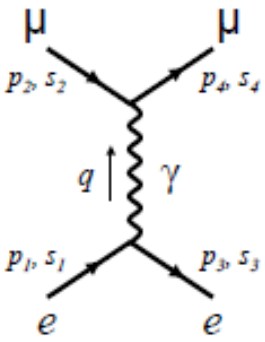
$$-e\bar{u}_f \gamma_\mu u_i e^{i(p_f - p_i)\cdot x}$$

Flux j

They're not as jibberish-y as they look

The basic unit jμ for Dirac rules

Electron-Muon Scattering cont'd



Propogator is $\frac{-ig_{\mu\nu}}{q^2}$. Applying the Feynman rules we obtain

[spin indices have been suppressed]

$$\int \left[\bar{u}(p_3)\, ig\gamma^{\mu}\, u(p_1)(2\pi)^4\delta^4(p_1-p_3-q)\right]\frac{-ig_{\mu\nu}}{q^2}\left[\bar{u}(p_4)\, ig\gamma^{\nu}\, u(p_2)(2\pi)^4\delta^4(p_2+q-p_4)\right]\frac{d^4q}{(2\pi)^4}$$

Use first δ-function for integration over d^4q. This leaves:

$$\frac{-ig^2}{(p_1-p_3)^2}\left[\bar{u}(p_3)\gamma^{\mu}u(p_1)\right]\left[\bar{u}(p_4)\gamma_{\mu}u(p_2)\right]\cancel{(2\pi)^4\delta^4(p_1+p_2-p_3-p_4)}$$

j^{μ}

Canceling the overall δ-function leaves us with -i$\mathcal{M}$:

$$\mathcal{M}=\frac{g^2}{(p_1-p_3)^2}\left[\bar{u}(p_3)\gamma^{\mu}u(p_1)\right]\left[\bar{u}(p_4)\gamma_{\mu}u(p_2)\right]$$

Think of the "vector" as flux

- Instead of jibberish : Flux = j_μ

MAIN TAKE AWAY (CLASS) ABOUT DIRAC STUFF! (makes simpler to think about)

Flux j

$$j_\mu^{fi} = -e\bar{\psi}_f \gamma_\mu \psi_i$$

$$= -e\bar{u}_f \gamma_\mu u_i e^{i(p_f - p_i)\cdot x}$$

Flux j

Also recall that generally a flux j = $\rho\ \vec{v} \propto \vec{p}$
for four-j j = (ρ, **j**)

3- reasons for making this association:

Flux Reason 1): "Simple Electro-dynamics (halzen and martin)

Rapidity problems solutions

$p = p_T \cosh\eta$

1) use exponential relations like e^y direct substitution of y. then algebra
2) a) direct differentiation of the $\tan(\theta/2)$ definition of pseudorap works but it's a little nicer if you use the $1+\cos\theta/1-\cos$ verson

b) $\sin\theta = pt/|p|$ → just rearrange $p = p_T\cosh\eta$; also E = p for m=0, also mT= pT etc. so you could use any of the similar y relations

c) see class

3&5) I apologize this one may have been a bit difficult –partially b.c. some of the problem needs more clarification, and also for first part because I forgot to emphasize more a few things in class I did mention but not enough emphasis:
dy = dp_z/E - for 3, proving this is whole thing—not too hard see slide after**, for 5,** this leads to best ideas on jacobian e.g. but first most straightforward but messiest, you could use relations for E,pz in terms of y or p and pz in terms of η sub these in opposite def: e.g. take d/dη of y= (- ½ ln ([sqrt(m^2+ptcosh$^2\eta$)-ptsinη]/[same w/ + where "-" was]) ugly in the end you did not need to simplify really, and just plot
but ~easy to prove relation w/ β using $dy = dp_z/E$
$Edy = dpz$ → $pz = pt \sinh\eta$

fixed/const

$\rightarrow dp_z/d\eta = p_T \cosh\eta \rightarrow \frac{E\,dy}{dp_z} \Rightarrow E\,dy = dp_z = p_T \cosh\eta\, d\eta$ (|p|)

$dy = \frac{p}{E} d\eta = \beta\, d\eta$

finally there is an alt form of y I also forgot emphasize → because pz/p = cosθ and $\beta_{||}$ from slides is $\beta\cos\theta$ ($p_{||}$ =pcos

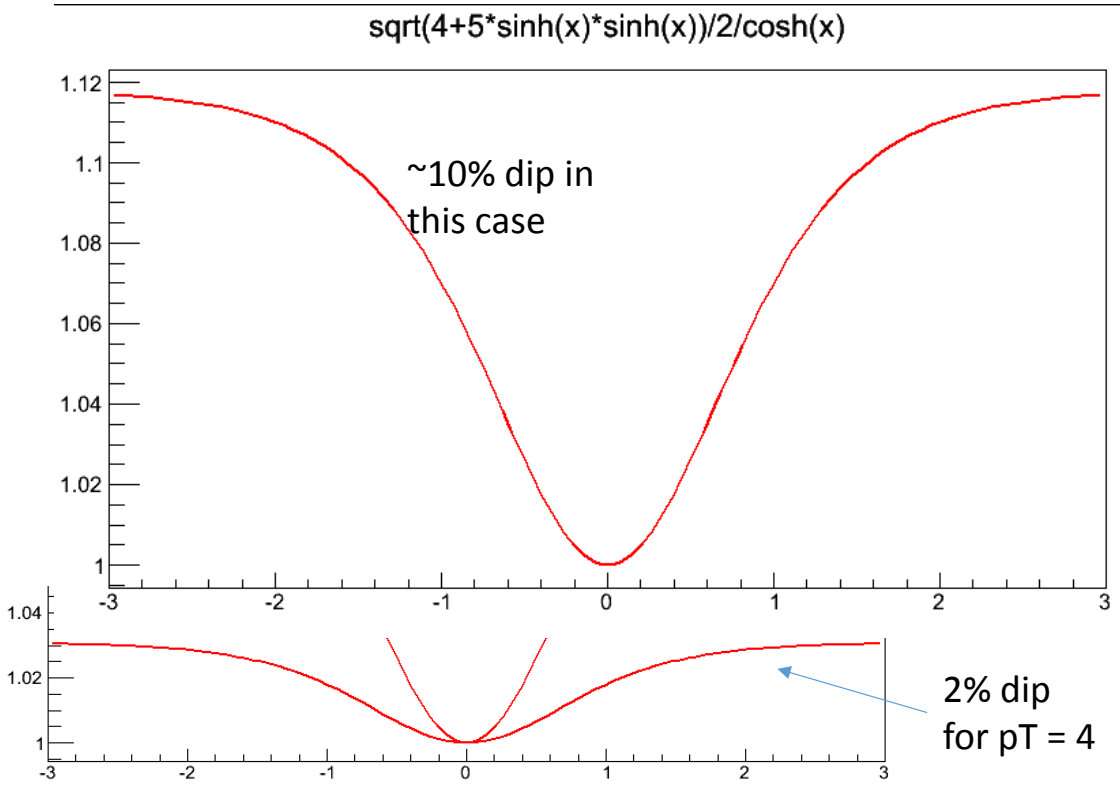
$\Rightarrow y = \frac{1}{2}\ln\left(\frac{1+\beta\cos\theta}{1-\beta\cos\theta}\right)$

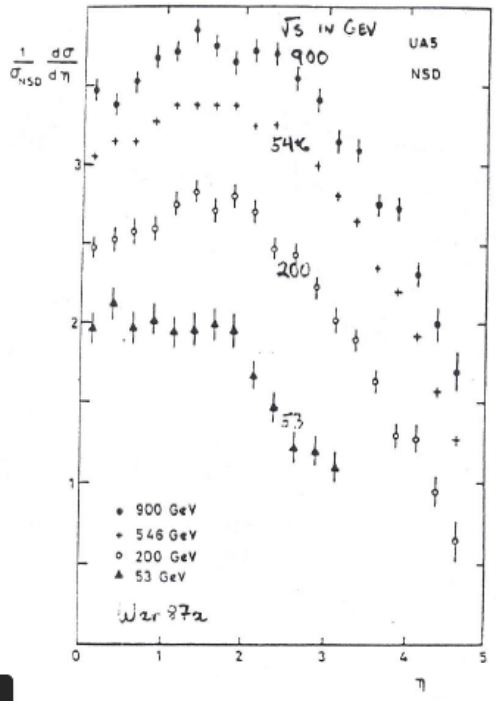
β in this is real beta, super similar form to eta can get $dy = \beta d\eta$ then with small approx. ratio-ing $d\eta/d\cos\theta$ with $dy/d\cos\theta$

Rapidity problems solutions

$$\beta(y) = \frac{\sqrt{p_T^2 + p_z^2}}{E} = \frac{p}{E} = \frac{\sqrt{p_T^2 + m_T^2 \sinh^2 y}}{m_T \cosh y}$$

- **5)b)** ie jacobian is this function. dN/dη = $\beta(y)$ dN/dy so plot $\beta(y)$
- But here's other big thing I'd should've clarified for this problem, for both part a) [this part] and part b) you ~need to specify pT to be ***something*** which you can pick—you should choose a relativistically large value (for which β ~1 even at $\theta = 90$degrees)—also in this case too as with part b obviously you need to specify the mass –the effect of the jacobian will be bigger with bigger mass (I just happen to know this) so let's go with proton mass
- Also choose pT = 2 GeV→ so mT^2 = pT^2 + m^2 ≈5





REAL WORLD (29)

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'Flat' only in a qualitative sense

Not especially depressim about $\eta = 0$.

Exercise: How much of this depressim results from difference between η and y?

Hints: 1.) Show $dy = \beta d\eta$

2.) Estimate correct assuming all particles π's

Rapidity problem solutions 5)c

- again just comparing y and eta, but setting y = n, can only determine ratio of $p_{||}/E$ (pz/E) …must specify m, AND $\mathbf{p_T}$ to fix p_z, |p|, and E
- from pz/p = cosθ (determine θ) → calc η

pt 2 GeV protons		**y=**	**y=**	**y=**
y = n --> =1/2 ln (1+p/E)/(1-pz/E) choose n=		**1**	**2**	**3**
e^2 = (1+p/E)/(1-pz/E)				
use my inversion trick				
pz/E = -(1-e^2)/(1+e^2) =		0.761594156	0.96402758	0.995054754
pick pt = 2 , m = 1				
(pz = 0.7616*(sqrt(pz^2+5)) [b.c. E sqrt(pz2+pt2+**m**)		8% diff	5%	3%
pz^2 = 0.7626^2(pz^2 + **5**)				
1-0.7616^2 * pz^2 = 0.76^2*5--> pz = sqrt(0.76^2/(1-0.7616^2)		2.627829756	8.109906417	22.40064933
cos thet= pz/p = pz/(sqrt(pz^2+**4**) [p sqrt(pz2+pt2) no m		0.79574634	0.970911555	0.996037934
eta = 0.5*ln(1+costh)/(…		1.086906674	2.107955204	3.111076884
pt 2 GeV PIONS		**1**	**2**	**3**
	0.018769	0.761594156	0.96402758	0.995054754
		0.2% diff	0.1%	< 0.1%
beta‖ = p‖//C = pcosthea/E				
betall/beta		2.356271067	7.271832506	20.08577677
		0.762391052	0.964197051	0.995079176
	3.090633162	1.001900235	2.002404275	3.002481469

Rapidity problem solutions 4

- Problem 4. time dilation factor = cosh y means prove γ = cosh y_{beam}
- any "situation" just meant "situation 1" in slides is OK here: $\beta_{||} = \beta$ in this case : b.c. we use the beam particles to define the special direction we will calc all other particles y w.r.t. so **their** (actual) boost parameters (β, γ) can be written in alt cosh/sinh way as defined in the slides.
- E = mT cosh y → mt = m (beam particle pT = 0)→ E= mcosh y→
- E/m = coshy → γ = cosh y ✓
- notice more so that the second relation $p_{||}$= $m_T$sinhy → "$p_{||}$→ p"

ie becomes p = msinhy take 1/E both sides →β = msinhy / E

$\gamma\beta$= sinhy (both γ & $\gamma\beta$ for the Lorentz trans)